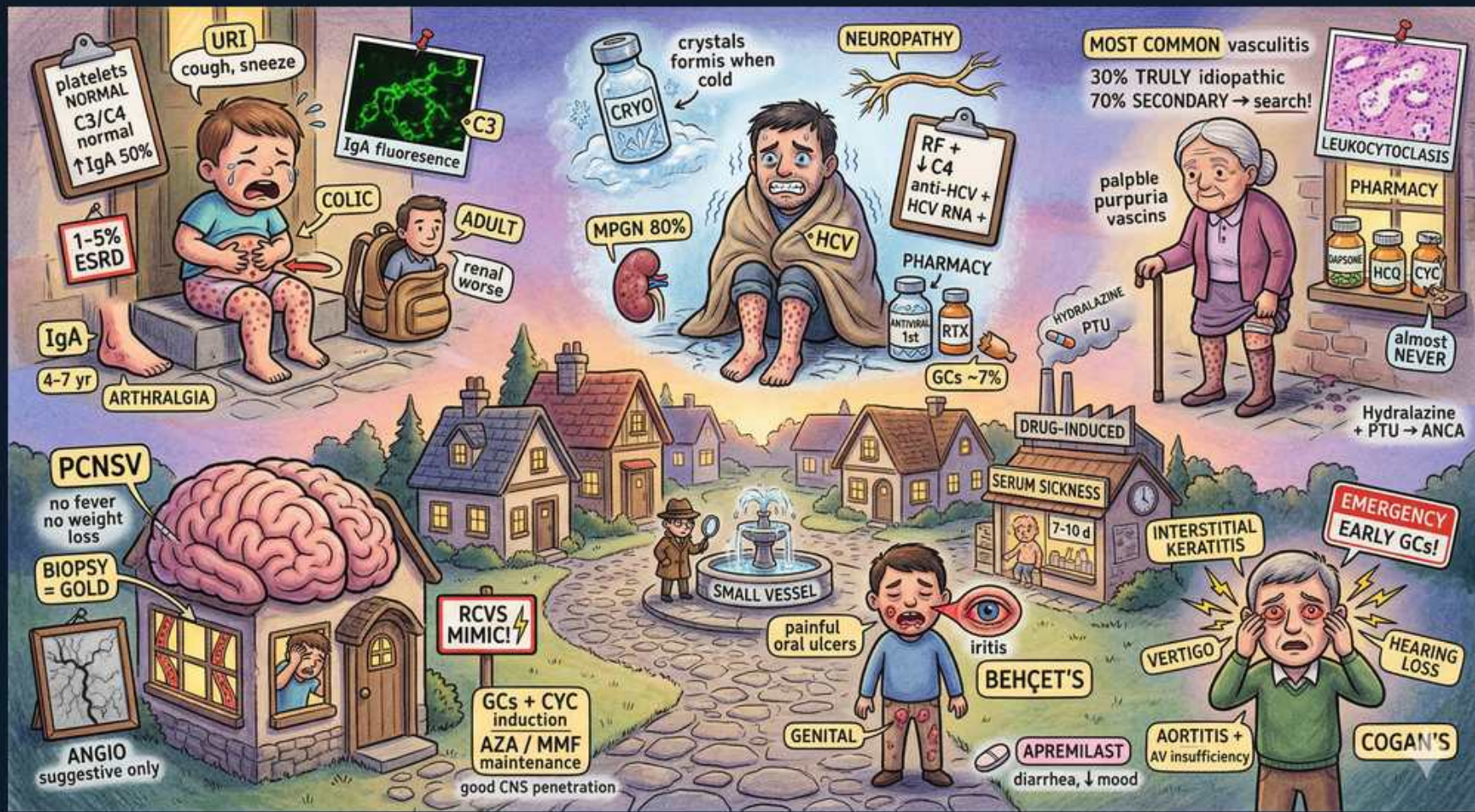


Vasculitis — Small-Vessel / Immune-Complex & Special Types



1. IgA / HSP child

WHAT YOU SEE

Child with purpura on legs, IgA, 4-7 yrs, URI cough/sneeze

WHAT IT ENCODES

IgA vasculitis (Henoch-Schönlein purpura) is the most common childhood vasculitis (peak ~4-7 yr), often 1-2 weeks after an upper respiratory infection. Classic tetrad: palpable purpura on the buttocks/lower extremities (dependent areas), arthralgia/arthritis, colicky abdominal pain, and renal disease (IgA nephropathy). It is immune-complex mediated with IgA deposition on biopsy.

BOARD TRAP

The purpura must be PALPABLE and in dependent/gravity-dependent areas. Non-palpable purpura suggests thrombocytopenia (ITP) or another cause, not vasculitis. Diagnosing IgA vasculitis with flat, non-palpable lesions or a low platelet count is the trap.

2. IgA labs — normal plt, ↑IgA

WHAT YOU SEE

platelets NORMAL, C3/C4 normal, IgA 50% tags

WHAT IT ENCODES

IgA vasculitis labs: platelet count is NORMAL (even elevated) and complement (C3/C4) is NORMAL — distinguishing it from thrombocytopenic purpura and from complement-consuming immune-complex vasculitides. Serum IgA is elevated in ~50%; skin/renal biopsy shows IgA-dominant immune deposits. Diagnosis is largely clinical in a classic pediatric presentation.

BOARD TRAP

Normal platelets + normal complement is the key discriminator: rules out ITP/TTP (low platelets) and lupus/cryoglobulinemia/post-strep (low complement). Selecting an answer with low complement or thrombocytopenia for IgA vasculitis is the trap.

3. IgA renal / ESRD

WHAT YOU SEE

1-5% ESRD, renal worse in adult tags

WHAT IT ENCODES

IgA vasculitis nephritis (histologically identical to IgA nephropathy) is usually self-limited in children, progressing to ESRD in only ~1-5%. Renal disease is more severe and more often chronic in ADULTS. Monitor urinalysis and BP for weeks after presentation; persistent proteinuria/hematuria or nephritic features warrant nephrology referral and possibly steroids/immunosuppression.

BOARD TRAP

Assuming benign self-resolution in an ADULT with IgA vasculitis is the trap — adult renal involvement is worse and may need aggressive therapy. Also, steroids relieve abdominal/joint pain but do NOT reliably prevent or treat the nephritis.

4. Cryoglobulinemic — HCV

WHAT YOU SEE

Man with leg purpura, CRYO crystals form when cold, HCV

WHAT IT ENCODES

Cryoglobulinemic vasculitis is most commonly driven by chronic hepatitis C (mixed type II/III cryoglobulins = polyclonal IgG + monoclonal IgM rheumatoid factor). Cryoglobulins precipitate in the cold, causing immune-complex small-vessel vasculitis: palpable purpura, arthralgia, peripheral neuropathy, and membranoproliferative glomerulonephritis (MPGN). The Meltzer triad is purpura + arthralgia + weakness.

BOARD TRAP

Treat the ROOT cause — direct-acting antivirals (DAAs) to cure HCV — rather than relying on immunosuppression alone. Giving long-term steroids/cyclophosphamide without antiviral therapy in HCV-cryoglobulinemia is the trap (rituximab is added for severe organ-threatening disease).

5. Cryo labs — RF, low C4

WHAT YOU SEE

RF + low C4, anti-HCV, HCV RNA, MPGN 80% tags

WHAT IT ENCODES

Type II cryoglobulinemia labs: positive rheumatoid factor (the monoclonal IgM has RF activity), characteristically LOW C4 with relatively preserved C3, positive anti-HCV and HCV RNA, and MPGN on renal biopsy in most with renal disease (~80%). Sample handling matters — blood must be kept warm (37°C) en route to the lab or cryoglobulins are falsely missed.

BOARD TRAP

A disproportionately LOW C4 with near-normal C3 plus positive RF is the cryoglobulinemic signature. Falsely 'negative' cryoglobulins because the sample cooled before processing is a classic trap — collect and transport at body temperature.

6. Cutaneous LCV

WHAT YOU SEE

MOST COMMON vasculitis, LEUKOCYTOCLASTIC, 30% idiopathic 70% secondary

WHAT IT ENCODES

Cutaneous leukocytoclastic (small-vessel) vasculitis is the most common vasculitis overall: skin-limited palpable purpura, biopsy showing neutrophilic infiltration with leukocytoclasia (nuclear debris) and fibrinoid necrosis of post-capillary venules. About 70% is secondary (drugs, infections, connective-tissue disease, malignancy) and ~30% idiopathic — so search for a trigger and confirm there is no systemic organ involvement.

BOARD TRAP

Skin-limited LCV rarely needs systemic immunosuppression — withdraw the offending drug/treat the trigger and use supportive care. Reflexively starting steroids/cyclophosphamide for isolated cutaneous LCV (or failing to search for the secondary cause/systemic involvement) is the trap.

7. Drug-induced — serum sickness

WHAT YOU SEE

DRUG-INDUCED house, SERUM SICKNESS panel

WHAT IT ENCODES

Drug-induced vasculitis / serum sickness is a skin-predominant, immune-complex small-vessel vasculitis appearing days to ~2 weeks after exposure (classically heterologous proteins/antitoxins, antibiotics like beta-lactams/sulfonamides, and monoclonals). Features: fever, urticarial rash/palpable purpura, arthralgia, and sometimes nephritis with low complement. It resolves on stopping the offending agent.

BOARD TRAP

The cornerstone is identifying and STOPPING the culprit drug — escalating immunosuppression instead of discontinuing the agent is the trap. Distinguish true serum sickness from a serum-sickness-LIKE reaction (e.g., to cefaclor), which lacks immune complexes/hypocomplementemia but is also managed by withdrawal.

8. Hydralazine/PTU → ANCA

WHAT YOU SEE

Hydralazine + PTU -> ANCA, almost NEVER pharmacy tags right

WHAT IT ENCODES

Drug-induced ANCA-associated vasculitis: hydralazine, propylthiouracil (PTU), minocycline, and cocaine adulterated with levamisole are the classic culprits. They often produce HIGH-titer p-ANCA/anti-MPO (and sometimes anti-PR3, anti-elastase, or multiple antigen positivity). Management is drug withdrawal ± a short course of immunosuppression for organ-threatening disease.

BOARD TRAP

Cocaine/levamisole-induced vasculitis closely MIMICS GPA (midline destructive lesions, ANCA positivity, retiform purpura with neutropenia/agranulocytosis) — immunosuppressing without taking a drug history and stopping the agent is the trap. Anti-human-neutrophil-elastase antibodies favor cocaine/levamisole over idiopathic GPA.

9. PCNSV brain

WHAT YOU SEE

Brick house brain labeled PCNSV, no fever, no weight loss

WHAT IT ENCODES

Primary CNS vasculitis (PCNSV/PACNS) is vasculitis confined to brain/spinal cord vessels: subacute headache, cognitive decline, and recurrent strokes/TIAs in multiple territories, WITHOUT systemic inflammation. Characteristically there is NO fever or weight loss and a NORMAL ESR/CRP. CSF often shows mild lymphocytic pleocytosis and elevated protein.

BOARD TRAP

Systemic features (fever, weight loss, high ESR) argue AGAINST primary CNS vasculitis and toward a SECONDARY cause (infection, malignancy, systemic vasculitis, drugs). Diagnosing PCNSV in a patient with prominent systemic inflammation is the trap — that picture demands a search for a secondary etiology.

10. PCNSV biopsy gold / angio

WHAT YOU SEE

BIOPSY=GOLD, ANGIO suggestive only, RCVS mimic tags

WHAT IT ENCODES

PCNSV diagnosis: brain (leptomeningeal/cortical) biopsy is the gold standard, though false negatives occur from patchy involvement; conventional/CT/MR angiography is only SUGGESTIVE (beading/stenoses are non-specific). The crucial mimic is RCVS (reversible cerebral vasoconstriction syndrome) — thunderclap headache, often post-partum or after vasoconstrictive drugs/SSRIs, with reversible 'string-of-beads' arteries.

BOARD TRAP

Angiographic 'beading' alone does NOT confirm PCNSV — RCVS produces identical findings but is non-inflammatory and resolves spontaneously (normal CSF, reversible on repeat angiography in ~12 weeks). Immunosuppressing RCVS as if it were PCNSV is the trap and can be harmful.

11. PCNSV treatment

WHAT YOU SEE

GCs + CYC induction, AZA/MMF, good CNS penetration tags

WHAT IT ENCODES

PCNSV treatment: induction with high-dose glucocorticoids PLUS cyclophosphamide (good CNS penetration), then maintenance with azathioprine or mycophenolate mofetil. Rituximab is used in refractory cases. Treatment is prolonged with imaging follow-up given relapse risk.

BOARD TRAP

Do NOT empirically immunosuppress before excluding RCVS and infection — treating a non-inflammatory mimic with cyclophosphamide is the trap. Confirm inflammation (biopsy/CSF) and rule out RCVS (repeat vascular imaging) before committing to immunosuppression.

12. Behçet — oral/genital ulcers

WHAT YOU SEE

Painful oral ulcers, GENITAL ulcers, iritis labeled BEHCET'S

WHAT IT ENCODES

Behçet disease is a variable-vessel vasculitis (can affect arteries AND veins of any size). Hallmarks: recurrent painful oral aphthous ulcers (required), recurrent genital ulcers, ocular disease (uveitis), skin lesions (erythema nodosum, papulopustular), and a positive pathergy test. It is associated with HLA-B51 and is most prevalent along the historic Silk Road.

BOARD TRAP

Behçet uveitis is frequently POSTERIOR or panuveitis with retinal vasculitis/hypopyon — sight-threatening, not merely benign anterior iritis. Treating Behçet eye disease as simple anterior uveitis (topical steroids alone) is the trap; it needs systemic immunosuppression/biologics. Pulmonary artery aneurysms are a unique, dangerous Behçet feature.

13. Behçet — apremilast

WHAT YOU SEE

APREMILAST, diarrhea/weight loss tag

WHAT IT ENCODES

Behçet treatment is organ-tailored: mucocutaneous disease responds to colchicine, topical steroids, and apremilast (a PDE4 inhibitor approved for oral ulcers; side effects include diarrhea, nausea, weight loss). Severe organ disease (eye, CNS, large-vessel/aneurysmal, GI) needs systemic glucocorticoids plus azathioprine, cyclophosphamide, or TNF inhibitors (infliximab/adalimumab); interferon-alpha is an option for refractory uveitis.

BOARD TRAP

Apremilast/colchicine handle mucocutaneous Behçet but are NOT adequate for sight-threatening uveitis, CNS, or vascular disease — using them as monotherapy for severe organ involvement is the trap. Escalate to TNF inhibitors/immunosuppression for major-organ Behçet.

14. Cogan syndrome

WHAT YOU SEE

Elder with VERTIGO, HEARING LOSS, INTERSTITIAL KERATITIS, AORTITIS labeled COGAN'S

WHAT IT ENCODES

Cogan syndrome is a rare autoimmune vasculitis defined by interstitial keratitis (red, painful, photophobic eye) PLUS audiovestibular dysfunction resembling Ménière's (vertigo, tinnitus, sensorineural hearing loss). It can also cause large-vessel disease — aortitis with aortic regurgitation. Treatment is high-dose glucocorticoids, with immunosuppressants/biologics for systemic or refractory disease.

BOARD TRAP

The sensorineural hearing loss is an EMERGENCY — prompt high-dose systemic steroids can preserve hearing, and delay risks permanent deafness. Treating only the eye (topical therapy) while observing the ear is the trap; also screen for aortitis/aortic regurgitation.